Storytelling and Finalizing the app
Activity Book

Create Visualizations with Qlik Sense

CVQS May 2023

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Activity 1: Working with snapshots

Summary:

- Explore and work with snapshots in a Qlik Sense app.
- Understand how to use the snapshot library and manage snapshots.

Qlik Sense app: ABC Sales Dashboard v2

Challenge

Using the **ABC Dashboard** sheet in the **ABC Sales Dashboard v2** app, create the following snapshots below using the listed charts and selections. Add the specified annotations as well:

- KPI Sales CYTD with no selections (annotation: "*CYTD Sales - No selections*")
- KPI Sales CYTD with EMEA division selected (annotation: "*CYTD Sales - EMEA*")
- Bar chart Sales CYTD - LYTD with EMEA division selected (annotation: "*Sales YTD Comparison - EMEA*")
- Pie chart with EMEA division selected (annotation: "*Sales share - EMEA*")
- Pie chart with EMEA division and 2023 year selected (annotation: "*Sales share - EMEA 2023*")
- Combo chart with EMEA division and Sportswear selected (annotation: "*Sales YTD trend - EMEA Sportswear*")

Note you can edit these annotations after creating them using the **Snapshot library**.

Hints

- In analysis mode, you can **Take snapshots** of different visualizations by doing a right-click on them and selecting **Take snapshot**.
- The existing selections are applied to the snapshot at the time the snapshot was taken. You can add **Annotations** to describe the details of the selection status.
- Snapshots can also be edited or removed from the **Snapshot library**.

Step-by-step guide

1. Open the **ABC Dashboard** sheet in the **ABC Sales Dashboard v2** app, and clear all selections, if necessary.
2. Right-click on the existing KPI *Sales CYTD* and take a snapshot. Add an annotation, *CYTD Sales - No selections*. Then select *EMEA* using the pie chart and take a new snapshot of the Sales CYTD KPI again, using the following annotation: *CYTD Sales - EMEA*.
3. Right-click on the existing Bar chart *Sales CYTD - LYTD* and take a snapshot with *Sales YTD Comparison - EMEA* as the annotation.
4. Right-click on the pie chart (donut) and take a snapshot using *Sales share - EMEA* for the annotation. Then select 2023 using the trend line chart and take a new snapshot of the pie chart. This time use the following annotation: *Sales share - EMEA 2023*.

5. Right-click on the pie chart and click on **Storytelling snapshots** > **Snapshot library**. From here you can view, edit, or delete the snapshots, if needed, by using the @ edit option. Compare with the solution image.
6. Close the **Snapshot library**. Clear the selection in the *Year* and select *Sportswear* using the pie chart.
7. Hover over the existing combo chart and take a snapshot with *Sales YTD trend - EMEA Sportswear* as the annotation.

Solution

Activity 2: Working with stories

Summary:

- Explore and work with stories in a Qlik Sense app.
- Create stories and include snapshots and other objects in slides.
- Understand the navigation and export options in stories.

Qlik Sense app: ABC Sales Dashboard v2

Dependencies: Before working in this activity, make sure you complete Activity 1.

Challenge

Create a story in the **ABC Sales Dashboard v2** app called *EMEA Report* using the snapshots you captured in the previous activity.

The story should focus on the sales analysis for the EMEA region and should include the following items:

- Three slides (use an embedded sheet in one of them)
- At least one **Text title** and one **Shape**
- **Snapshots** and **Effects** (use these three effects: **Lowest value**, **Highest value**, and **Any value**)

Play the story and delete one of the slides.

Examine export options.

Bonus point:

- Add a thumbnail to the story using one of the images in the Default library.

Hints

- Stories can be created and edited from the **Storytelling** menu in an app.
- **Snapshots**, **Text**, **Shapes**, and **Images** can be added to a slide when editing the story.
- Effects can be applied on existing snapshots in the slide by dragging them from the **Effects library** onto the snapshot. For the **Any value** effect, you can select the specific value you want to highlight.
- A thumbnail image and a description can be added to the story when editing the info panel of the story.
- When editing a story, you can use the **Global Menu** to explore the export options.

Step-by-step guide

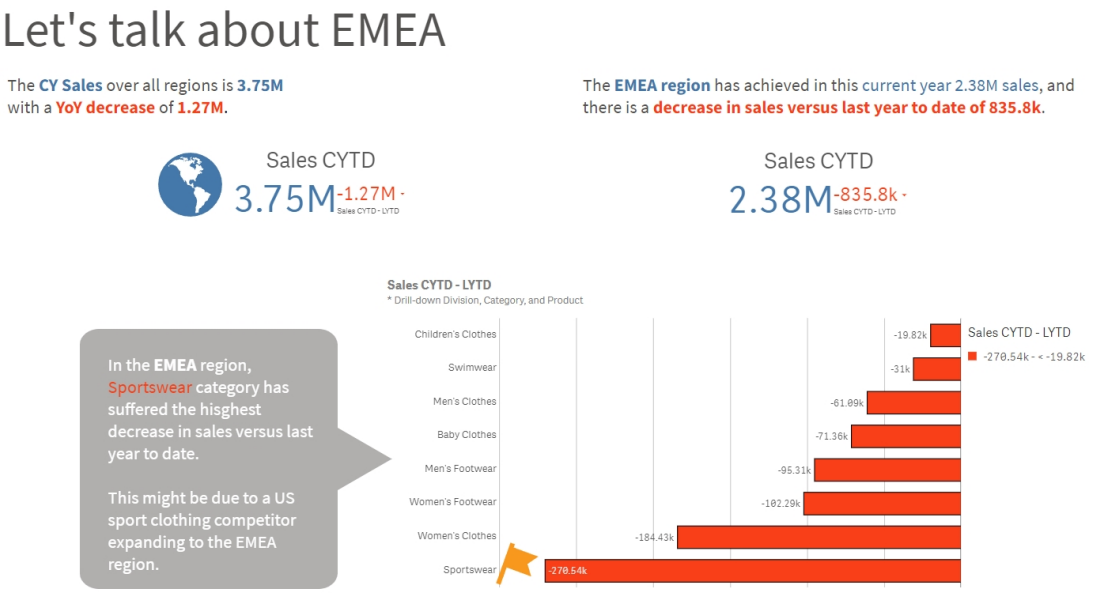
1. In the **ABC Sales Dashboard v2** app, create a new story called *EMEA Report* using the **Storytelling** option from the toolbar.

   ### First slide

2. Open the story and drag a **Title** onto the slide from the **Text objects** section. Then double-click in it to add the text, *Let's talk about EMEA*.

3. Using the **Snapshot library**, drag the two KPI snapshots into the slide. Then drag a **Paragraph** from the **Text objects** section and add some text to explain the numbers.
4. Using the **Shapes library**, drag a **World** shape next to the KPI for all regions. Set the color to dark blue.
5. Add the bar chart snapshot onto the slide. The bar chart is displayed in the slide as an image.
6. Resize the image, using the unlock padlock for free resizing, then click @ edit to hide the legend and click **Done**.
7. Using the **Shapes library**, drag a **Flag** shape on top of the bar chart pointing at the top bar in red. Resize the flag and select a color for it. (For instance, use a red color)
8. Drag another **Shape** and **Paragraph** next to the bar chart image to put some context in the image: "*In the EMEA region, Sportswear category has suffered the hisghest decrease in sales versus last year to date. This might be due to a US sport clothing competitor expanding to the EMEA region*".
9. Highlight the name of the product category. Format part of the text to different color or font if necessary.
10. Right-click on the bar chart image and select **Go to Source**. This allows the user to review more details from the story in the app.
11. Navigate back to the story. The first slide is completed and it will be similar to the example below:

Slide 1 - Example



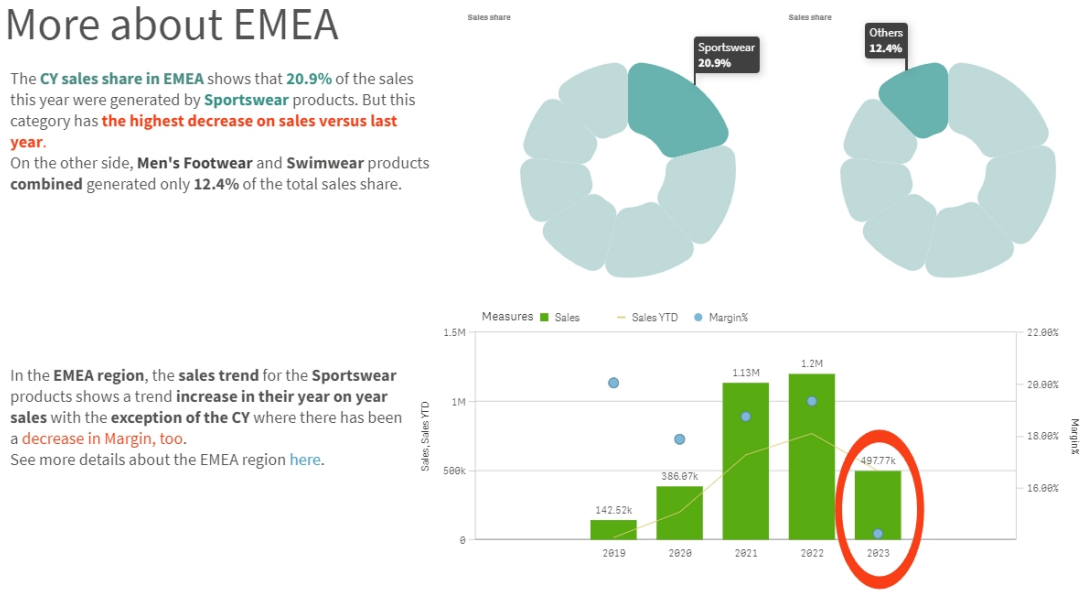
Second slide - Duplicate a slide in a story

12. Duplicate the current slide to create a second slide.
13. On the second slide, change the title to read the following: *More about EMEA*.
14. Delete all the other images, paragraphs, and shapes in the second slide.
15. Drag a snapshot containing the pie chart with the **Sales share - EMEA 2023** annotation twice onto the second slide. Now you have the same snapshot repeated twice on the second slide.

16. Using the **Effects library**, drag the **Highest value** on the first pie chart snapshot. The visualization changes, highlighting the product category within *EMEA 2023* with the highest sales share, *Sportswear*.
17. Using the **Effects library**, drag the **Lowest value** onto the second pie chart snapshot. The visualization changes, highlighting the category within *EMEA 2023* with the lowest sales share. Please note if you have "Others" in the chart, go to the source and click on others to see what categories are the lowest. In this case, *Men's Footwear* and *Swimwear*. Using the **Effects library**, drag the **Any value**, click the effects icon in the snapshot, and select *Others* from the drop-down.
18. Drag the combo chart snapshot to the bottom right of the slide and edit the snapshot to remove titles. Then use the **Shapes library** to add a red circle over the year 2023 in the combo chart.
19. Add a paragraph to explain the selections in the chart and provide some details. Include also a link to the *EMEA* bookmark.

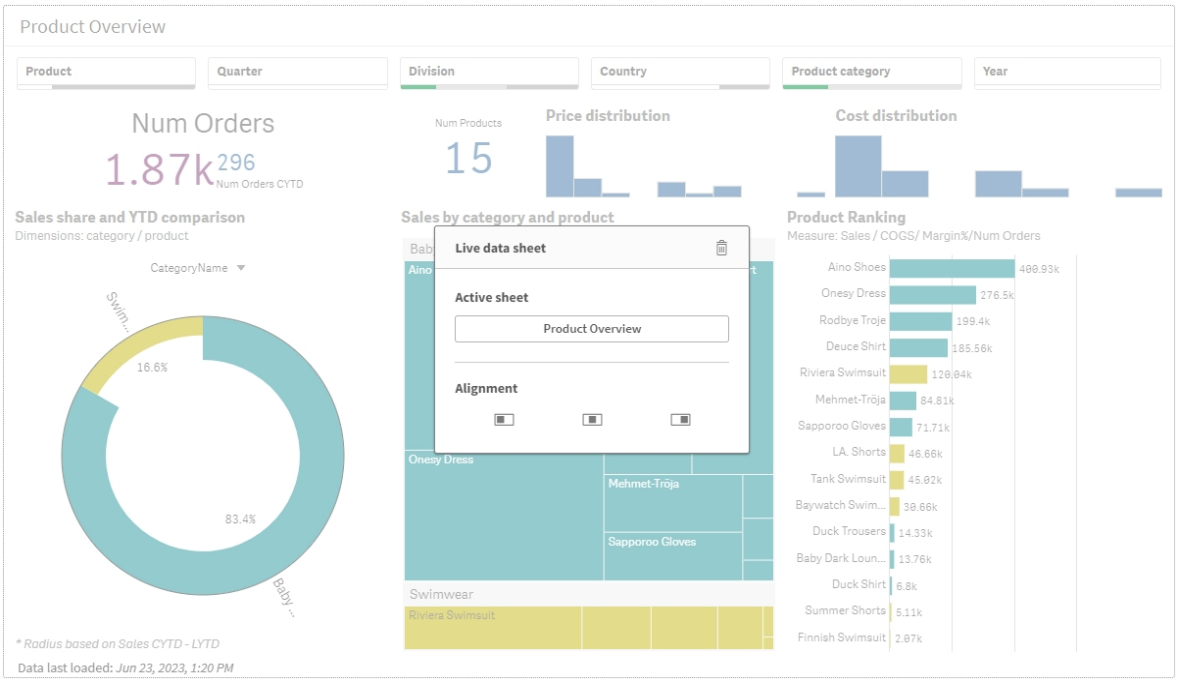
> **Note:** You can include links to URLs or bookmarks within the text in the paragraph. Select the text you want to link and use the **é Create link** icon. Then you can enter a URL or select an existing Bookmark. For example, choose *EMEA*.

Slide 2 - Example



Third slide - Embedded sheet

20. In the **Analyze** > **Sheet** section, navigate to the **Product overview** sheet and select *Baby Clothes* and *Swimwear* as Product categories and the *EMEA* division, then navigate back to the story (using the Storytelling section) and add the **Product overview** sheet from the **Sheet Library**. A third slide is created with the **Product overview** sheet embedded.

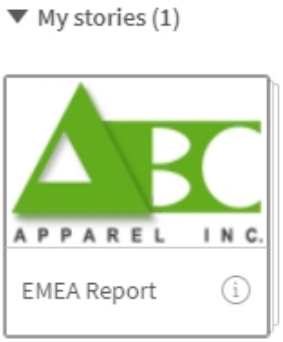


21. Select the first slide in the story and click **Play story**. Then use the **Actions** button at the bottom of the screen to advance to the next sheet or go back to the previous sheet. Notice how you can make selections in the third slide using the filters from the **Product overview** sheet or charts and hover over the bars to see the tool tips.
22. Close the Play view to return to editing the story, right-click on the second sheet, and then **Delete** it.
23. From the global menu, examine the export options.

Bonus point:

Add a thumbnail:

24. From the **Stories** menu, hover over the **EMEA Report** story and open the **] Details** in edit mode.
25. Click the image to **Change thumbnail** and insert an image from the default library.



Note: Stories can be exported to PDF format. Use the **Global Menu** on the top toolbar to see the export and download options and other actions. Another format available in Qlik Sense on Windows editions is PowerPoint.

Activity 3: Prepare the app for delivery

Summary:

- Understand how to review and prepare the app for handing over to consumers.
- Work with master items, descriptions, and tags.
- Add items to Favorites and monitor charts in the Qlik Cloud hub.

Qlik Sense app: ABC Sales Dashboard v2

Challenge

Prepare the **ABC Sales Dashboard v2** app to hand over to business users. To accomplish that, do the following:

- Delete any unused sheets and clean up any unused visualizations or objects.
- Make sure every sheet is clear and all the visualizations are easy to understand. You can always include a sheet description, and use the titles, subtitles, and footnotes in visualizations. Showing details in charts can also help users to understand the visualization.
- Add more information in descriptions and tags in the existing master measures and dimensions, if needed. For instance, add a description in the COGS master measure.
- If you are working with Qlik Sense cloud editions, monitor some charts in the Hub.
- Show details in the following charts, create master visualizations based on them, and include descriptions and tags:
  - Combo chart in the **Dashboard** sheet
  - Scatter plot in the **Product analysis** sheet

Hints

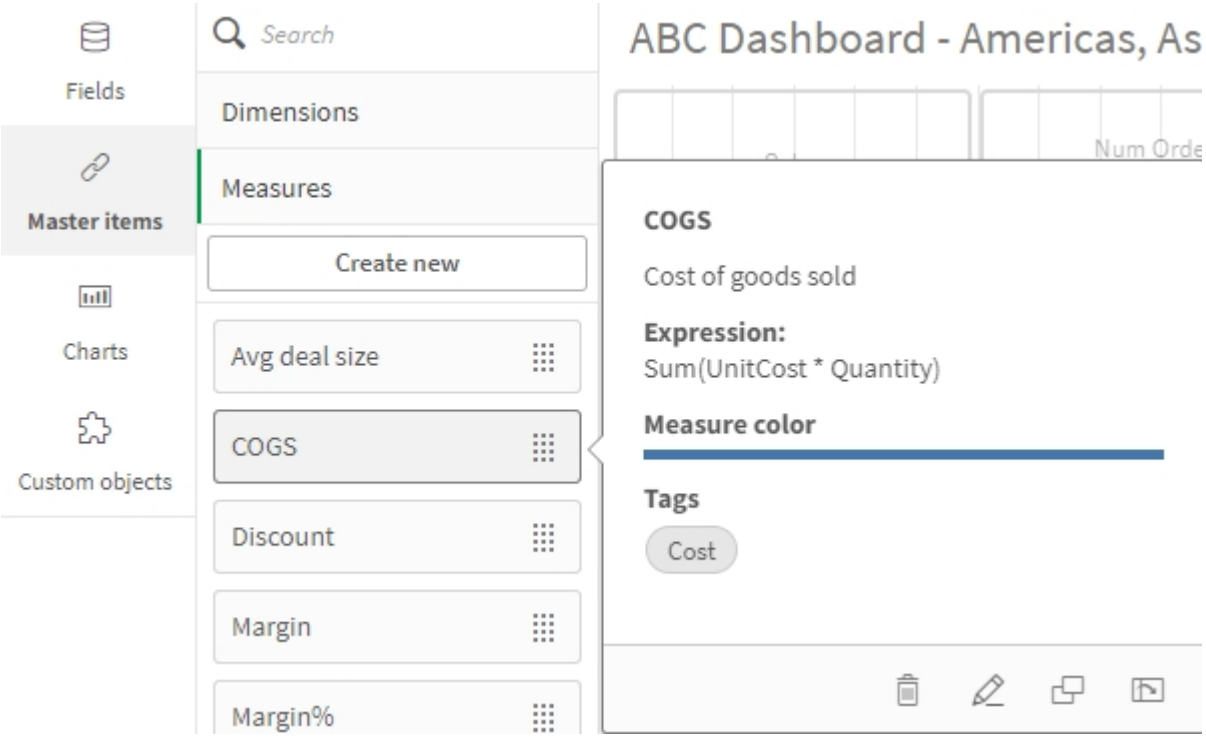
- Add comments and labels to master dimensions and measures, using the **Description** and **Tag** areas when editing the master items.
- The **Show details** option can be found in the visualization properties panel, under the **Appearance** > **General** section.
- Right-click on a visualization and select **Add to master items** to create a master visualization.

Step-by-step guide

1. Open the **ABC Dashboard** sheet in the **ABC Sales Dashboard v2** app, in edit mode.

Add comments and information

2. From the **Assets** panel, under **Master items** > **Measures**, edit the **COGS** master measure.
3. Type *Cost of goods sold* as **Description** and add *Cost* as one of the **Tags** for this measure. Then **Save** the changes.



Monitor visualizations in the Hub (only for Qlik Sense cloud editions)

4. In analysis mode, select *2023*, then right-click on the *Sales* KPI and select **Monitor in hub**. Add *Total Sales 2023* as name to the visualization and click **Monitor**.
5. Clear selections and right-click on the *Sales and YTD Sales* trend, then select **Monitor in hub**. Add *Sales and Margin trend* as name and monitor the visualization.
6. Go to the Hub (under **Home > Your charts** section) and review the charts you are monitoring. Click on the *Total Sales 2023* KPI to go into the app again.

Add the app to Favorites (only for Qlik Sense cloud editions)

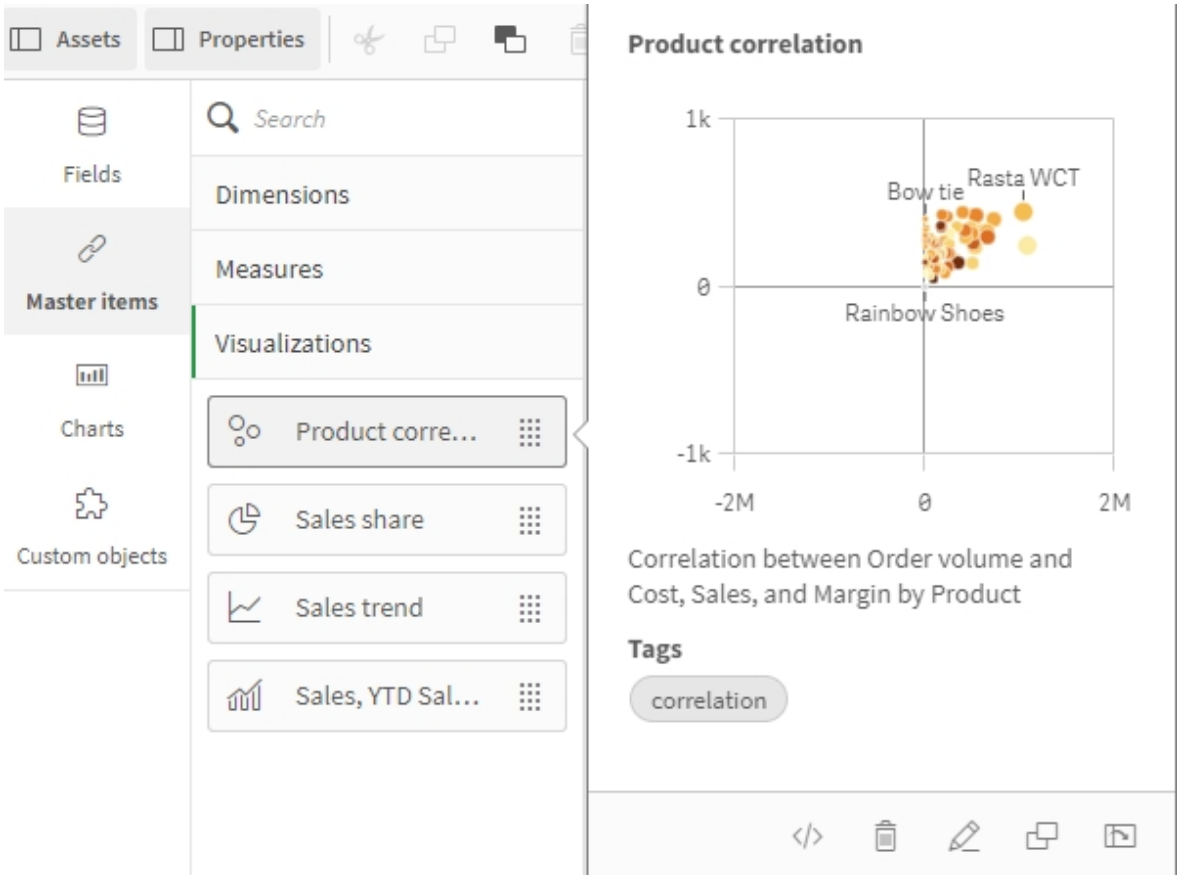
7. As it is an app that we will share with others, it could be convenient to save it in **Favorites**. This way you will be able to find the app easily in the Hub. Click on the **Star** icon when hovering over the app tile.
8. Go to **Home** in the Hub, and check that the app is under the Favorites section.

Display the object details and create master visualizations

9. In edit mode, select the existing combo chart and using the properties panel, under the **Appearance** > **General** section, set the **Show details** to **Show**.
10. Right click on the combo chart and select **Add to master items**. Use *Sales, YTD Sales, and Margin% trend* as **Name**. Note that you can add description and tags.
11. Click **Add**. The visualization is added to the master items list.
12. In the **Product analysis** sheet, select the scatter plot, and use the properties panel to set the **Show details** to **Show**.
13. Right-click on the scatter plot and select **Add to master items**.
14. Type an informative name, description, and tags and click **Add**. Notice how the master visualizations are now linked visualizations and the properties panels are not available.

Note: The master visualization can be edited from the **Assets panel** > **Master items**. You will need to unlink the visualizations in order to make changes to it.

Solution



(Optional) Activity 4: Explore your app on tablets and small-screen devices

Summary:

- Understand how to prepare the app for small screen devices.
- Explore and simulate user access with different screen sizes.

Qlik Sense app: ABC Sales Dashboard v2

Challenge

In this activity, you will be given an opportunity to test how your app displays on small-screen devices.

Google Chrome's Developer Tools enable you to go through popular device screens without leaving the browser environment. All you have to do is select a device from the Console Emulation tab.

You also have full control of the emulation parameters, such as screen resolution, touch emulation, user agent, sensors, and many more.

To perform your test, complete the following steps:

1. With one of the **ABC Sales Dashboard v2** app sheets open, launch the **Developer tools** on Google Chrome.
2. Use the Toggle device mode to select different mobile types and review the app.

Hints

- Open a sheet in analysis mode in order to use the developer tools.

Step-by-step guide

1. Open the **Product overview** sheet, in the **ABC Sales Dashboard v2** app. Make sure you are not in edit mode.
2. Open the **Developer tools** and under the **Toggle Device Mode** or **Emulation**, select the device model by changing the *Responsive* option in the drop-down menu.

   Qlik Sense Desktop:

   a. Access the Qlik Sense Hub in a Google Chrome browser with the URL: **http://-localhost:4848/hub/my/work**

   b. Open the app in the Hub.

   c. Follow instructions for all Qlik Sense editions.

   All Qlik Sense editions:

   a. Open the **Google Chrome** menu at the top-right of your browser window.

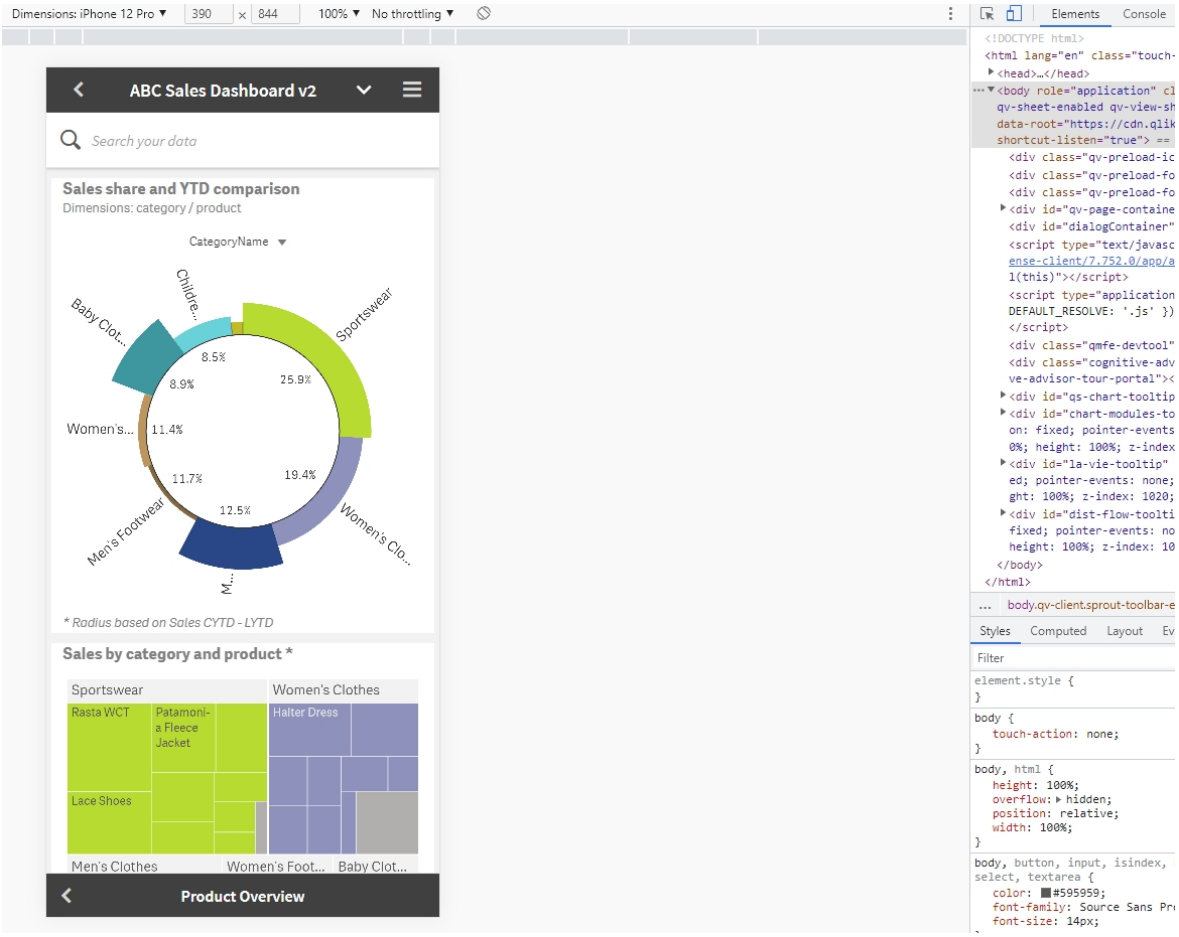
   b. Select **More tools** > **Developer tools** (or press **Ctrl+Shift+I**).

   c. Press the **Toggle Device Mode** button (or press **Ctrl+Shift+M**).

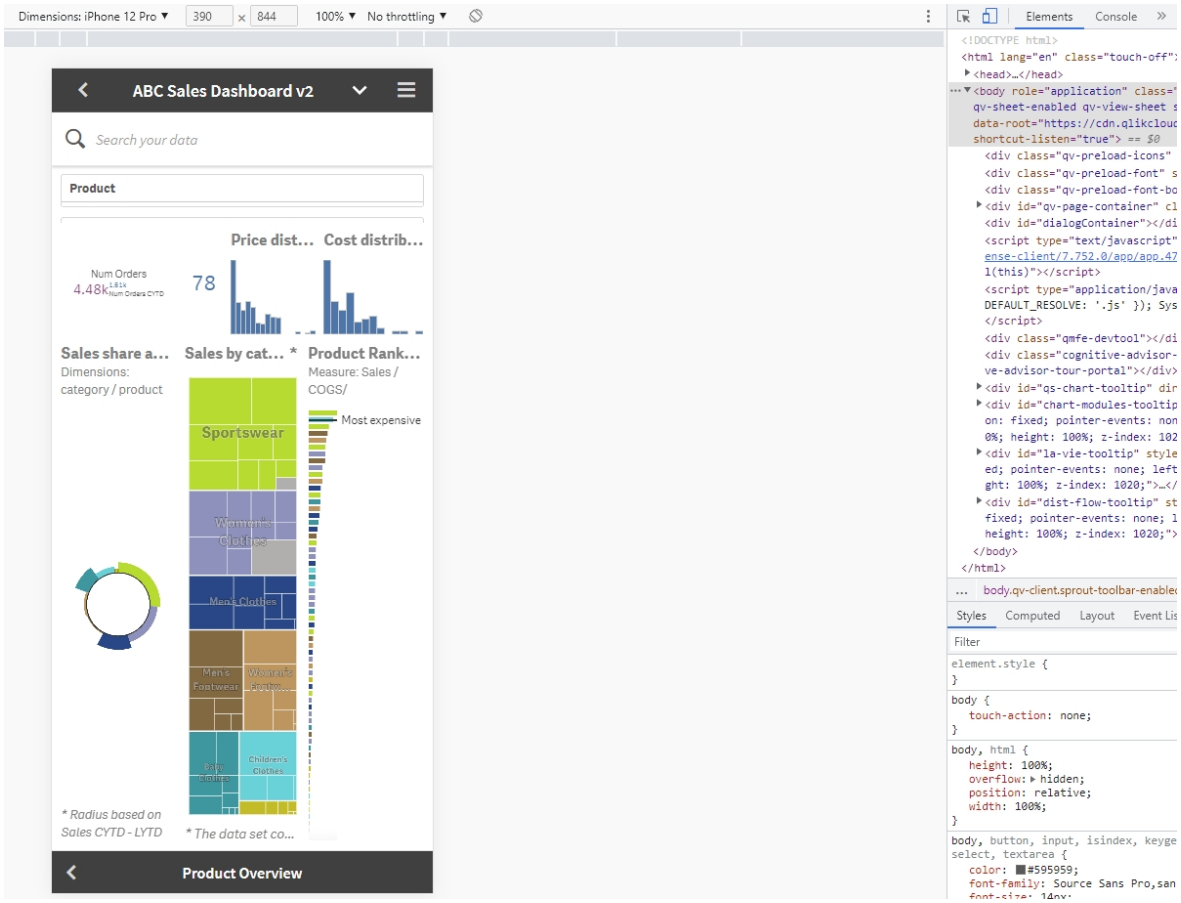
3. Notice how the **Product overview** sheet is now displayed, as if you were using the selected device. In the **Emulation** pane, select a different model from the Device menu. If necessary, press **F5** to reload the page.
4. Explore the app by opening sheets and interacting with objects.
5. Close the developer tools to go back to the standard view. Then turn the **Edit** mode on in the sheet.
6. In the sheet properties, change the **Small screen layout** property to **Grid view**. Then turn the edit mode off.
7. Open the **Developer tools** again and explore the differences.

Solution

List View



Grid View


Dimensions: iPhone 12 Pro
390
844
100%
No throttling
Elements
Console
ABC Sales Dashboard v2
Search your data
Product
Price dist...
Cost distrib...
Num Orders
4.48k
78
Sales share a...
Dimensions:
category / product
Sales by cat... *
Product Rank...
Measure: Sales /
COGS/
Sportswear
Women's Clothes
Men's Clothes
Men's Footwear
Baby Clothes
Children's Clothes
Most expensive
* Radius based on
Sales CYTD - LYTD
* The data set co...
Product Overview
Styles
Computed
Layout
Filter
element.style {
}
body {
touch-action: none;
}
body, html {
height: 100%;
overflow: hidden;
position: relative;
width: 100%;
}

Activity 5: Style your app

Summary:

- Understand how to review and prepare the app for handing over to consumers.
- Work with master items, descriptions, and tags.

Qlik Sense app: ABC Sales Dashboard v2

Challenge

In this activity, you set a thumbnail image for the app and apply a style to all the sheets.

For the thumbnail image, choose any image you like from the media library or from the ABC images folder.

When styling the app, make sure you:

- Review the available themes. You can also include other existing themes or build your own theme.
- Apply a background color for the sheet titles using a color range.
- Change the font color in the sheet titles.
- Add a logo or an image at the right of the sheet titles.

Feel free to chose your own colors and images. You can also add new images to the library if needed.

Bonus point:

- Apply a custom theme (available under the Student Files > Themes)
- Take a screenshot of each sheet in the app in analysis mode and use it as a thumbnail for each sheet

Hints

- A thumbnail can be added to the app using the app **Edit** @ panel.
- In the **App options** V panel, you can select a standard or custom theme, apply title background color, and add a logo to all sheets. The custom themes need to be added to Qlik Sense before you can apply them.

  Cloud editions of Qlik Sense: In the management console (*https://<your tenant address>/console*), under the **Themes** section, you can add new themes.

  Qlik Sense Enterprise on Windows: Use the **QMC > Extensions** section to add any new custom themes.

  Qlik Sense Desktop: Create a folder called *Themes* under: **..\Users\<UserName>\My Documents\Qlik\Sense\Extensions**. Then for each theme, make sure you include the theme files (JSON and the QEXT files).

- New images can also be included in the media library.

Qlik Sense Desktop: Copy images manually into the location where the product was installed. By default, the location is under the users' **Documents\Qlik\Sense\Content\Default** folder.

Step-by-step guide

1. In the **App Overview**, in the **ABC Sales Dashboard v2** app, click on the @ icon to access the app edit panel. In there you can change the name of the app or add a description.
2. Click on the thumbnail image. The media library dialog appears.
3. Choose the *ABC Sales Dashboard.png* image from the **In app** section. Then click **Insert**. Note that you can preview the images when you select the image names.

   Note: If the image is not in the In app list, you can import it as a new image in the app. You can add new images to the Media library using the **Upload media** option.

4. Confirm the changes and open the **ABC Dashboard** sheet.
5. Show the app information by clicking on the app name at the top. The app information and the thumbnail are now displayed on the top.
6. Click the **App options** V button. Then under **Appearance**, review the different options for **App theme**.
7. Under the **App capabilities**, make sure both **Insight Advisor in hub** and **Insight Advisor in this app** options are turned on.
8. Under the **Sheet title styling**, select the left and right colors for the background and a font color compatible with the background color.
9. Click the 0 icon to select a logo and choose an image from the media library. You can use the ABC company logo.
10. Make sure the **Image alignment** is set to right.

    Note: Under **Appearance** and **Sheet title styling** respectively, you can turn off **Show toolbar**, as well as **Show sheet header**. If you want other users to see more clearly and with fewer options the sheets of your app when they access it, they are some options.

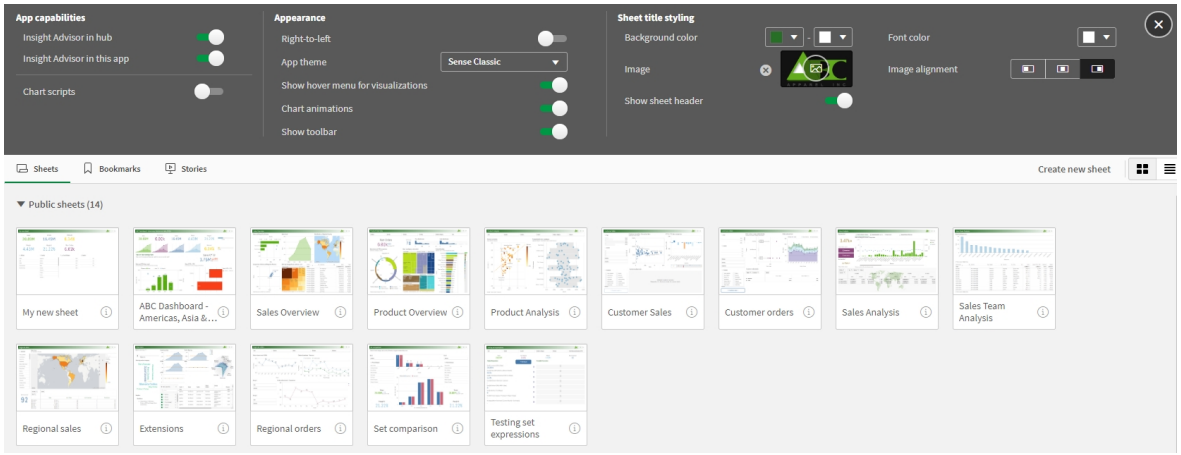
11. Close the app options and navigate through the sheets in the app to see the results.

    Bonus point:

12. Include the custom themes provided in the Student files, and review how the app looks when applying them using the **App options > Appearance > App theme**. If you have not included these custom themes before, make sure you follow these steps first:
    a. If using Qlik Sense Business or other Saas editions, access the Management console (*https://<your tenant address>/console*) and add the new themes in the **Themes** section.
    b. If using Qlik Sense Enterprise on Windows, open the **Qlik Management Console**, and under **Extensions**, import each zip file corresponding with each theme.
    c. If using Qlik Sense Desktop, navigate to **..\Users\<UserName>\My Documents\Qlik\Sense\Extensions**, and create a folder called Themes. Then unzip all the custom themes in this folder.

13. In analysis mode, navigate through the sheets taking a screenshot of each sheet and save it in a location you choose. If you are working with Qlik Sense Desktop, use the default folder for images (**C:\Users\<user>\Documents\Qlik\Sense\Content\Default**).
14. Open the **ABC Dashboard** sheet in edit mode and click on the sheet name. Then in the sheet properties panel, click the 0 icon to select a **Thumbnail**.
15. In the **Media library**, upload the image if needed, then select the corresponding image and click **Insert**.
16. Navigate through each sheet in edit mode and add the corresponding thumbnail. Then navigate to the **App overview** and review the result.

Example



Activity 6: Set alerts, share and interact with others

Summary:

- Understand how to set conditions in the app to trigger alerts base on the data.
- Publish or move the app to streams or spaces.
- Learn how to create subscription reports based on visualizations.
- Work with others using notes.

Qlik Sense app: ABC Sales Dashboard v2

Challenge

In the **ABC Sales Dashboard v2** app, create an alert that gets trigger when Margin% is under 15% based on the current year for each division in America.

Make some of the sheets public in the app if you are working with Cloud editions of Qlik Sense and share it with others using any of these options:

- Move the app to a shared space (in Qlik Cloud)
- Publish the app to a managed space (in Qlik Cloud) or to a stream (in Qlik Client-Managed)

Qlik Sense on Windows: Notes and subscription reports are not supported.

Qlik Sense Desktop: This activity is not supported.

Bonus point:

Create a report about the ABC Sales and send it to your email.

Hints

- When creating the alert, apply the *Division* dimension and the *Americas* bookmark to define the data set. Then compare the Margin% with the 0.15 value.
- In Cloud editions of Qlik Sense, make sheets and stories public before publishing or moving the app to a space.
- You can create alerts based on measures in the app and apply specific dimensions and selections.
  - In Qlik Cloud, right-click on a visualization to set an alert. You can manage and monitor your alerts in the Qlik Cloud Management console. Note **Data alerts** must be enabled in the Settings section of the Management console.
  - In Qlik Client-Managed, use the Qlik Alerting extension to enable users to create alerts. You can create and manage system and data alerts from the Qlik Alerting portal.
- You can subscribe to reports about visualizations and sheets in an app. Note: **Subscriptions** must be enabled in the Settings section of the Management console, under the Email sharing section. An email server must be also configured for users to receive reports by email.
- You can create notes to collaborate with others and share your insights. Note: **Notes** must be enabled in the Settings section of the Management console, under Feature control section.
- Reports can be created using a **Qlik App Application Automation** and the **Qlik Reporting** connector available in there.

Step-by-step guide

1. In the **ABC Sales Dashboard v2** app, open the **ABC Dashboard** sheet in edit mode. Make sure all selections are cleared.

> **Note**: Steps differ between Qlik Client-Managed and Qlik Cloud. Follow the corresponding steps based on your environment.

Qlik Cloud

Create alert

1. Go to the **Alerts** section in the Qlik Cloud Hub. From here you can enable or disable alerts and manage them. Now you will create your own alert in the app.
2. In the **ABC Dashboard** sheet, right-click on the *Margin%* KPI and create a new alert.
3. Under **Title**, enter *Margin% lower than 15%*. *Margin%* is already added. Note, you could add additional measures.
4. Then select *Division* in the **Show by dimension** drop-down and the *Americas* bookmark under **Selections**. Note how the Data preview displays a table with two rows.
5. Click **Next** and set the condition selecting *Margin %*, **Less than**, and **Value** in the drop-downs. Then enter *0.15* in the **Value** input box. Note how the Preview alert section displays only the rows where the condition is met. If no rows meet the condition the alert is not triggered.
6. Click **Next** and set the notification frequency to trigger the alert weekly. Then click **Create**.
7. Note how the alert is displayed in the hub. Click on the notification message to see the details of the alert. In the **Alert** panel, you can review the alert details, view the history of the notifications for the alert, check recipients and navigate to the app.

Publish the app or moving the app to a space

8. In the **App overview**, under **Sheets**, right-click in some of the sheets under the My sheets area, then select **Make Public**.
9. Repeat the same for **Bookmarks** and **Stories** in the app.
10. In the Cloud hub, under **Catalog > Personal** area, click on the 3 dots button in the app you want to share and select **Publish** to share the app in a Managed space. Alternatively you can select **Move** if you want to share it in a Shared space.
11. Select the **Space** where the app is going to be published or moved. When you publish the app you can change the app title, description, and add tags. You can only publish apps to a managed space. If you don't have managed spaces in your Cloud environment you can use the **Move** option instead and move the app to a shared space.
12. Note, if you move the app to a shared space, the app is no longer in your personal area, while if you publish the app into a managed space, the app remains in your personal area as well for further development. Discuss other differences between managed and shared spaces. Your instructor will demonstrate how to manage user access to each space and the user rights available.

Create subscription report

13. The instructor will access the Cloud Management console and show you the **Subscriptions** section. From here the tenant administrator can enable or disable alerts and manage them. Now you will create your own subscription report in the app.

14. In the **ABC Dashboard** sheet, right-click on the *Top 10 selling products* bar chart and create a new subscription, and leave **Title** as it is.
15. Choose *Weekly* on **Schedule options**.
16. Note that you could add other users as recipients of the report if you want. Then click **Subscribe**.
17. Now the subscription is displayed in the hub. Click on the notification message to see the details. In the **Subscriptions** panel, you can review the details, the history of the reports for the subscription, check recipients and navigate to the app.

Create note

18. In the **ABC Dashboard** sheet, right-click on the *Sales share* pie chart and create a new note with the snapshot of the chart. Click on **Create note with snapshot**.
19. Type "*EMEA share is higher than Americas's*" as **Title**.
20. Write some text on the note. In case more users have access to the space where you have created the note (you can see the name of the space on the bottom of the Notes panel), you can type @ to mention them. They will receive a notification and will be able to participate on editing the note.

> **Note**: When a user is added to a space, whether directly as an individual user or indirectly as a member of a group, they will have **Can view** permissions for any notes created in the space.

21. In case you want to modify permissions of other users over you note, click on icon.
22. Then go to the **Hub > Catalog**, filter **Types** by *Notes* and check that the note is on the corresponding space.

Bonus point:

23. From the Hub, click on **Add new** > **New automation**.
24. Select **Reporting** under the **Categories** section and choose the **Send a simple multi-page report by email** template. Hover over the template tile and click the**Use template** button. Note that a series of blocks are already placed into the automation scheme.
25. Leave the **Start** block as it is.
26. Click on **Variable - recipients** and, on the right panel, write the email where you want to receive the report in the second gray box under the **Operations on recipients**.
27. Then, click on the **Create report** tile and select the app from the scroll bar. If you select **do lookup**, you will see the list of all the apps you have access to, and you can select it easily.
28. Go to **Send Mail** > **Connection**. Click the button **Connect your Mail**(in case you already have a connection configured, you can use it or create a new one). We will use a Gmail account. In order to ensure that the report can be delivered correctly, you must create an App password in your Gmail account. Follow this instructions to get it.
29. Then, complete the connection configuration, that will look like this:
    - Host: smtp.gmail.com
    - Port: 587
    - Security: starttls
    - Sender_Email_Address: your Gmail address

- Username: leave blank
- Password: the app password that you have just generated

30. Click **Run** on the top of the page. When finishes, you will receive the report in your email.

Qlik Client-Managed

Create alert

1. In a new Chrome browser tab, go to the **Qlik Alerting** portal: *https://qse:4552/#/*
2. If credentials are required, enter *Student* as name and *Training1* as password. From this panel you can monitor your alerts or create new ones.
3. Under **Admin > User management > Users**, review the **Email** column set for the users. Here you can change the email address of your user (*student*). Click the 3 dots in the student row and select **Edit**. Then under **Email ID**, enter your personal email and click **Save**.
4. Go back into the **Alerts** section to monitor the data alerts. You, as administrator, could create new alerts from here, but for this activity we will use the Qlik alerting extension to demonstrate how users can create their own alerts.
5. Go back to the Qlik Sense hub, and in the app, edit the *ABC dashboard* sheet and make some free space in the sheet.
6. From the **Assets panel > Custom objects > Extensions**, drag the **Qlik Alerting** extension onto the free space in the sheet.
7. In the extension properties panel, under **Qlik Alerting server**, replace *localhost* with *qse*. Then under **Style**, turn the **Show icon** property to **On**. This new button allows users to create their data alerts.
8. In analysis mode, click the **Create alert** button to trigger the alert creation wizard.
9. Set an alert name: *Margin% lower than 15%* and click **Next**.
10. From the **Measures** drop-down, select *Margin%*. Then click **Add measure**.
11. From the **Dimension (optional)** select *Division* and click **Next**.
12. In the conditions drop-down select *Margin%*, **Less than (<)**, and **Manual value**. Then enter *0.15* in the **Value** input box and click **Next**.
13. Under **Selections / Bookmarks** select the *Americas* bookmark, under **Frequency** select **Only once**, and for **Channels** select **Email**. Then click **Create alert**.
14. A message is displayed to confirm the alert is created. Click **Close**.
15. Navigate to the Qlik Alerting portal in the other browser tab, refresh the page and review the data alerts. You can trigger the new alert using the 3 dots button and selecting **Trigger**. Then select **Confirm**.
16. Check your email inbox to see if you have an email based on the alert. If you don't see the email, make sure you review your spam mail. Note this email is the one set for the *student* user you edited in step 3.

Publishing the app to a stream

17. Under **Personal > Work** area in the hub, right-click on the app and select **Publish**.
18. Select the *ABC Apps* stream. You can leave the name as is or give a new name to the app. Note if the app already exists in the stream you also have the option to replace the existing one.
19. Click **Publish**. Note how the app has been moved from your personal area to the *ABC Apps* stream.

20. Select the stream and open the published app. Note all the sheets are public and you cannot edit them. You can still create new sheets or duplicate existing sheets in this app.